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1 GLM Overview

1.1 Example

Normal Regression: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$: $\epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Logistic Regression: $y_i \stackrel{\text{ind}}{\sim} \text{Bern}(p_i)$: $\text{logit}(p_i) = \beta_0 + \beta_1 x_i$, $\log(\frac{p_i}{1-p_i})$

Differences:

- Support of y_i : $y_i \in \mathbb{R}$ vs. $y_i \in \{0, 1\}$
- Link functions: $g(\mu_i) = \mu_i$ vs. $g(p_i) = \log(\frac{p_i}{1-p_i})$
- Likelihoods: $y_i \stackrel{\text{ind}}{\sim} N(\mu_i, \sigma^2)$ vs. $y_i \stackrel{\text{ind}}{\sim} \text{Bern}(p_i)$

Similarities:

- Univariate models
- Exponential families
- Independently distributed
- Linear (or otherwise systematic) relationship with a function of $E(y_i)$
- Have link functions
- Assuming that y_i is random
- β 's are unknown

1.2 GLMs

Generalized Linear Model:

- Random component: $Y_i \sim f(\mu_i)$
- Link component: $g(\mu_i) = \eta_i$
- Systematic component: $\eta_i = x_i^\top \beta$

We generally use GLMs for the following:

- Inference: understand relationships
- Prediction: behavior of specific x 's

1.3 Poisson Regression

$Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, $\log(\mu_i) = x_i^\top \beta$

Assumptions:

- Linearity
- Independence

- Poisson distribution on response var

1.4 Exponential Family Calculations

With the Exponential Family log-likelihood written as: $l(\theta; y) = w(\theta)t(y) + \ln(a(\theta)) + \ln(b(y))$ where $\ln(a(\theta)) = c(\theta)$ and $\ln(b(y)) = d(y)$:

$$Score = w'(\theta)t(y) + c'(\theta)$$

$$\mathbb{E}[t(y)] = -\frac{c'(\theta)}{w'(\theta)}$$

$$Var(t(y)) = \frac{w''(\theta)c'(\theta) - w'(\theta)c''(\theta)}{(w'(\theta))^3}$$

Example: Poisson: $f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!} = \exp\{y \ln(\lambda) - \lambda - \ln(y!)\}$

In this case, $c(\lambda) = -\lambda$ and $w(\lambda) = \ln(\lambda)$

Sufficient Statistic: $t(y) = y$

$$w'(\lambda) = \frac{1}{\lambda}, w''(\lambda) = -\frac{1}{\lambda^2}, c'(\lambda) = -1, c''(\lambda) = 0$$

$$\text{Mean: } \mathbb{E}[t(y)] = \mathbb{E}[y] = -\frac{c'(\lambda)}{w'(\lambda)} = -\frac{-1}{\frac{1}{\lambda}} = \lambda$$

$$\text{Variance: } Var(t(y)) = Var(y) = \frac{w''(\lambda)c'(\lambda) - w'(\lambda)c''(\lambda)}{(w'(\lambda))^3} = \frac{(-1)(-\frac{1}{\lambda^2}) - 0}{\frac{1}{\lambda^3}} = \frac{\frac{1}{\lambda^2}}{\frac{1}{\lambda^3}} = \lambda$$

$$\text{Score: } w'(\theta)t(y) + c'(\theta) = (\frac{1}{\lambda})y + (-1). \text{ Verify } \mathbb{E}[Score] = 0 : \mathbb{E}[\frac{1}{\lambda}y - 1] = \frac{1}{\lambda} \mathbb{E}[y] - 1 = \frac{\lambda}{\lambda} - 1 = 0$$

1.5 Link Function

$$\eta_i = g(\mu_i)$$

Requirements (all with respect to the mean μ , not x):

- One-to-One ($g(a) = g(b) \implies a = b$)
- Monotone (Monotone increasing means $a > b \implies g(a) > g(b)$)
- Continuous and differentiable ($\lim_{\mu \rightarrow c} g(\mu) = g(c)$)

$$\text{Canonical Link: } g * (w(\theta_i)) = w(\theta_i)$$

Example,

$$\text{Binomial: } w(\pi_i) = \log(\frac{\pi_i}{1-\pi_i}) \implies g(\pi_i) = \log(\frac{\pi_i}{1-\pi_i}), g * (w(\pi_i)) = w(\pi_i)$$

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With U as the score function, and $\mathbb{J}$ as the variance:

$$U \sim N(\mathbf{0}, \mathbb{J})$$

$$\mathbb{J}^{-1/2}U \sim N(0, 1)$$

$$(\mathbb{J}^{-1/2}U)^\top \mathbb{J}^{-1/2}U = U^\top \mathbb{J}^{-1}U \sim \chi^2(k)$$

3 Prediction Intervals

Using regular process for confint on predictions of using the variance of $\frac{X^T X}{g^{-1}(\hat{\mu})}$ provides too large of an interval. The better method is to get interval on $\hat{\eta}$ then transform bounds to $\hat{\mu}$:

$\hat{\eta} = x\hat{\beta} \sim N(x\beta, XCov(\hat{\beta})X^T)$ Use that variance to get $SE(\hat{\eta})$

$$\hat{\eta} \pm 1.96SE(\hat{\eta}) = [\hat{\eta}_1, \hat{\eta}_2], \hat{\mu}_2 = g^{-1}(\hat{\eta}_2)$$

4 Residuals

Perason Residuals: $r_{p_i} = \frac{y_i - \hat{\mu}_i}{\sqrt{Var(Y_i)}}$

Examples:

Poisson: $\frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$

Logit: $\frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$

Anscombe: R doesn't really even touch

Deviance: uses the individual observation log-likelihood

Working: Similar to the Z statistic (difference from mean divided by SE)

5 Picking Link Functions

There are lots of ways to pick link functions:

- Interpretation
- AIC/BIC
- Specificity
- etc.

In some cases, the link function doesn't make much of a difference. However, sometimes it does.